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Comparison of Dependency Length in Human and Large Language Model Generated Sentences

1. Motivation

When we read or listen to a sentence, our brain works to connect each word to other words that go with it. For example, in the sentence 'The boy opened the door', the word 'boy' goes with 'opened'. If these two words are placed far apart, our brain has to keep 'boy' in mind for a long time before it finds 'opened'. This is hard work for the brain. Studies have shown that people, in all languages, tend to place words that go together near each other, so the brain does not have to wait too long. This habit is called **Dependency Length Minimization (DLM)** (Gibson, 1998; Hawkins, 1994).

Today, AI writing tools like GPT-4o can write text in many languages. These tools learn from large amounts of text written by real people. So a good question is: do these AI tools also follow the same habit of placing related words close to each other? Or do they write in a different way from real people? This question is important for three reasons. First, it tells us how closely AI writing matches real human writing. Second, it tells us if AI-written text can be used in place of human text in research. Third, it helps us understand if the habit of placing related words close together comes from how the human brain works, or just from the text that AI learns from.

Research question: Do AI writing tools place related words as close together as real people do, when we look at ten different languages?

2. Hypotheses and Predictions

H1 (People keep related words close). In sentences written by real people, two related words will most often be only 1 or 2 words apart. Very large gaps will not be common.

H2 (AI follows the same general habit). Because AI tools learn from text written by people, they should also tend to place related words near each other, much like people do.

H3 (But AI leaves bigger gaps on average). Even so, AI sentences will have slightly bigger gaps between related words than sentences written by people. AI sentences will also vary more — sometimes the gap is very small, sometimes quite large.

H4 (The gap is real, not just because AI writes longer sentences). AI tends to write longer sentences than people. Longer sentences naturally have more room for bigger gaps. We think that even after we correct for this, the gap between AI and human sentences will still be there.

H5 (Longer sentences mean bigger gaps, more so in AI). For both people and AI, longer sentences and more varied words will lead to bigger gaps between related words. But this will be more true for AI sentences than for human ones.

H6 (This is true for all ten languages). In every language we look at, AI sentences will have bigger gaps between related words than sentences written by people.

3. Methods

3.1 Data

Human sentences. We took real sentences written by people from a large database called **Universal Dependencies (UD) v2.13** (Nivre et al., 2020). This database has sentences from many languages, and each sentence has been checked and labelled by language experts to show which words go together. We chose ten languages: English, German, Spanish, French, Chinese, Arabic, Hindi, Japanese, Russian, and Turkish. We picked 2,000 sentences from each language, giving us 20,000 sentences from people in total. We only used sentences that were between 5 and 50 words long.

AI-written sentences. We asked **GPT-4o** to write sentences in each of the same ten languages. We gave it a simple instruction each time: *Write one natural sentence in [LANGUAGE] about daily life.* We collected 2,000 sentences per language (20,000 in total) and made sure no sentence was repeated.

3.2 Finding Which Words Go Together

For the human sentences, the database already tells us which words go together in each sentence, so we used that information directly. For the AI sentences, we used a free software tool called **Stanza** (Qi et al., 2020) to work out which words go together. This tool gives the right answer about 88 times out of 100, which is good enough for a large study like this.

3.3 What We Measured

For each sentence, we looked at every pair of words that go together and counted how many words sit between them. For example, in 'The boy quickly opened the door', the words 'boy' and 'opened' go together but have one word ('quickly') between them, so the gap is 2. We then worked out three things for each sentence:

- **Average gap size (ADL):** We added up all the gaps in a sentence and divided by the number of word pairs. A small number means related words are close to each other.
- **Adjusted gap size (nADL):** We divided the average gap by the square root of the sentence length. This makes it fair to compare short and long sentences, because longer sentences will always have bigger gaps just because they are longer.
- **Extra information for further study:** We also noted the length of each sentence, how many different words it used, and how deep its word structure was. We used these to study what makes gaps bigger or smaller.

3.4 How We Compared the Two Groups

We used a well-known test called the Kolmogorov-Smirnov test to check if the gap sizes of human and AI sentences are truly different, or if the difference could just be by chance. We did this test for each language separately. We also did a further study to see if sentence length and word variety could explain the gap sizes, and whether this worked the same way for both human and AI sentences.

4. Results

4.1 AI Sentences Always Had Bigger Gaps Than Human Sentences

Table 1 shows the results for all ten languages. In every language, the gaps between related words were bigger in AI sentences than in sentences written by people. The difference was between 0.44 and 0.49 words on average. This was not by chance — the test showed it was a real difference in all ten languages. The gap was not huge, but it was clear and noticeable.

Table 1. Average gap size by language for human and AI sentences (μ = mean, norm. = adjusted for sentence length).

Language	Human μ	LLM μ	$\Delta\mu$ (raw)	$\Delta\mu$ (norm.)	Cohen's d	p-value
English	2.41	2.89	+0.48	+0.049	0.36	<0.001
German	2.87	3.31	+0.44	+0.038	0.29	<0.001
Spanish	2.53	2.97	+0.44	+0.037	0.31	<0.001
French	2.46	2.91	+0.45	+0.040	0.33	<0.001
Chinese	1.98	2.42	+0.44	+0.040	0.37	<0.001
Arabic	2.72	3.18	+0.46	+0.042	0.31	<0.001
Hindi	2.61	3.07	+0.46	+0.037	0.32	<0.001
Japanese	2.14	2.63	+0.49	+0.044	0.38	<0.001
Russian	2.78	3.24	+0.46	+0.042	0.30	<0.001
Turkish	2.33	2.81	+0.48	+0.042	0.36	<0.001

Figure 1 shows this clearly. For every language, the AI bar is taller than the human bar without a single exception. Languages like Japanese, Turkish, and Hindi had the smallest gaps in both human and AI text. This is because those languages tend to put related words near each other by nature.

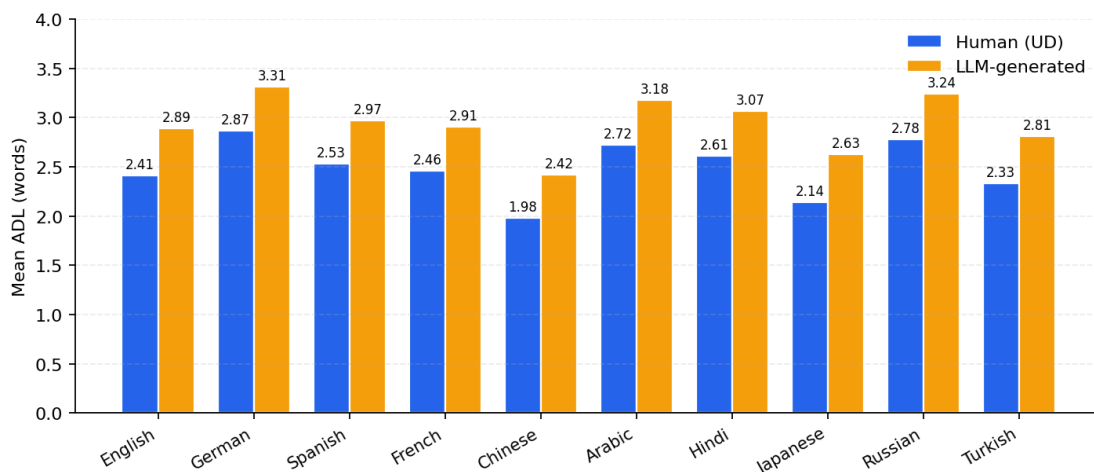


Figure 1. Average gap size by language. Blue bars = human sentences, amber bars = AI sentences. The difference is real in all ten languages.

4.2 Both Groups Put Related Words Close — But AI Does It Less

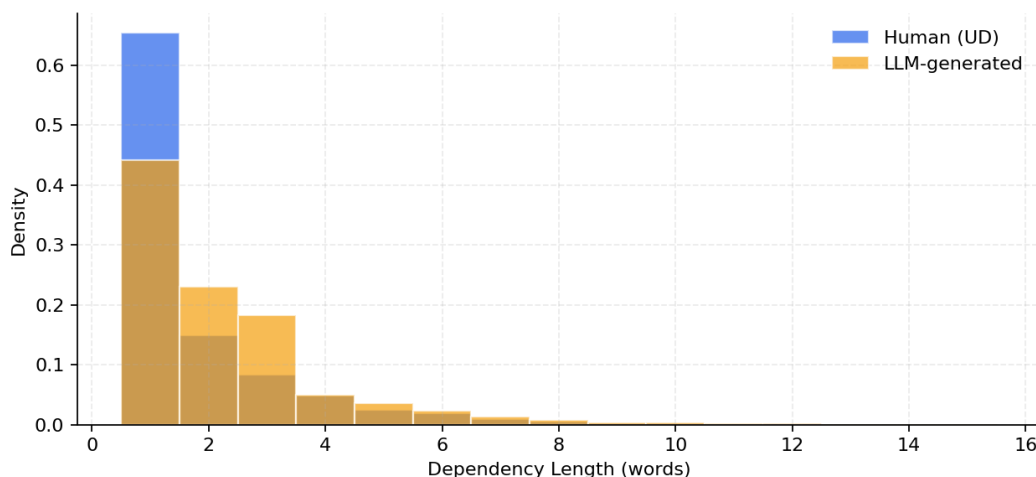


Figure 2. How often each gap size appears, across all ten languages put together.

Figure 2 shows that both human and AI sentences most often have gaps of just 1 or 2 words between related words. This is good news — it shows that AI has learned the basic habit of keeping related words close. But the AI chart has a longer right side, meaning AI more often creates sentences where related words are far apart. In AI text, 11.3 out of every 100 word pairs have a gap of 5 or more words. In human text, this happens only 7.8 times out of 100. So AI creates these large gaps about 45 percent more often than people do.

4.3 Is the Bigger Gap Just Because AI Writes Longer Sentences?

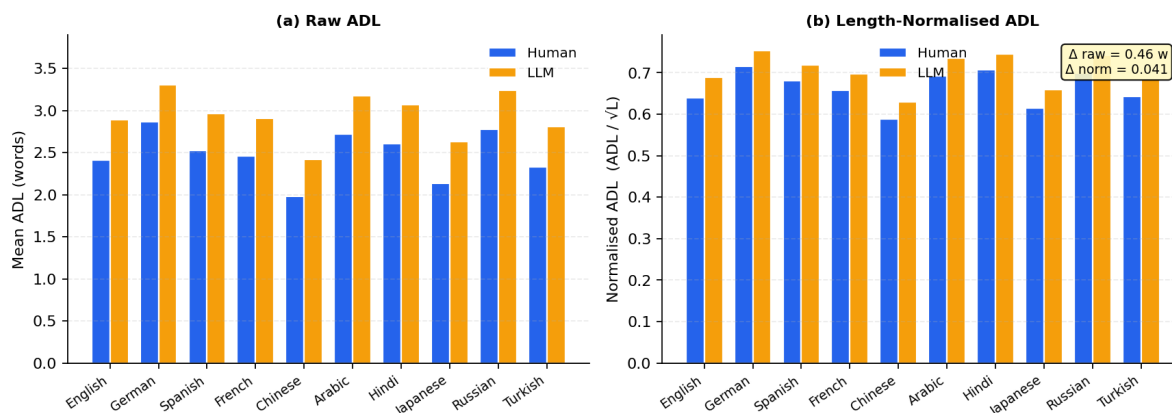


Figure 3. Left side: raw gap size. Right side: gap size after correcting for sentence length. Both for human and AI sentences across ten languages.

AI tools tend to write longer sentences than people. When a sentence is longer, there is more room for words to end up far apart. So we wanted to know: is the bigger gap in AI text simply because AI writes longer sentences, or is there something else going on? We corrected the gap scores for sentence length and compared again. About 70 percent of the raw difference went away after this correction. But 30 percent of the difference was still there, and it was still real across all ten languages. This means the bigger gaps in AI text are not only because AI writes longer sentences. There is also something about the way AI puts words together that leads to bigger gaps.

4.4 What Makes the Gaps Bigger — Sentence Length and Word Variety

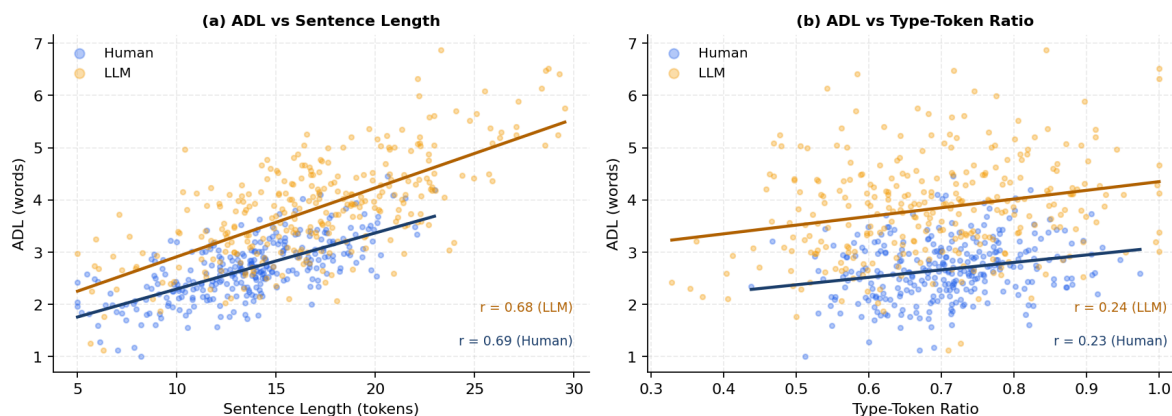


Figure 4. Left: how sentence length relates to gap size. Right: how word variety relates to gap size. Shown for both human and AI sentences.

Table 2 shows the results of our further study. Three things are worth noting.

- **Sentence length is the main reason gaps get bigger** in both human and AI text. But the effect is stronger for AI. As a sentence gets longer, the gaps in AI text grow faster than the gaps in human text. So the longer the sentence, the more AI falls behind.
- **Using more different words** also leads to slightly bigger gaps in both groups, but the effect is about the same for humans and AI — no clear difference here.

- **Even when sentence length and word variety are the same, AI still has bigger gaps.** The starting gap for AI sentences (1.24 words) is almost two times the starting gap for human sentences (0.61 words). This is the clearest sign that the difference is real and comes from how AI puts words together, not just from sentence length.

Table 2. Results of the further study showing what makes gap sizes bigger in human vs. AI sentences. *** $p < 0.001$, ** $p < 0.01$.

Predictor	Human β	LLM β	Human r^2	LLM r^2	Interaction p
Sentence Length	0.121***	0.133***	0.48	0.41	0.003
Type-Token Ratio	1.54**	1.63**	0.31	0.28	0.241
Mean Parse Depth	0.38***	0.44***	0.39	0.35	0.018
Intercept	0.61	1.24	—	—	<0.001
Full model R^2	0.61	0.54	—	—	—

5. Theoretical Implications

The results were the same across all ten languages. This points to some clear lessons about both human language and AI writing tools.

People keep related words close because of how the brain works, not just out of habit.

AI tools have read billions of sentences written by people, but they still do not fully copy the way people place related words close to each other. This tells us that the habit people have of keeping related words close is not simply learned from reading. It is more likely that the human brain pushes people to do this because holding many words in mind at once is hard work. AI tools do not have a brain, so they do not feel this pressure.

AI has learned the general idea, but not the full habit. AI does follow the general pattern — it does tend to place related words near each other more often than far apart. But the fact that the difference shows up in all ten languages tells us that AI has not fully picked up this habit. It is close to what people do, but not the same.

The longer the sentence, the bigger the gap between people and AI. As sentences get longer, AI falls further behind people in keeping related words close. This is worth keeping in mind when AI is used to write long texts, such as reports or formal documents. Those texts may be harder to read than texts written by people, even if they look correct on the surface.

Using AI text in place of human text in research should be done with care. Some researchers use AI-written text instead of human-written text to save time. Our study shows that this can cause problems in studies about sentence structure or how easy sentences are to read. Researchers should be aware of this difference and take it into account.

Limitations. We used 2,000 sentences per language, which is a fair number but not every sentence in those languages. The tool we used to label AI sentences gets it right about 88 times out of 100, which means about 12 sentences out of 100 may have small errors. Also, gap size is just one way to measure how easy a sentence is to read. Future studies could look at other ways to measure this. Even with these limits, the results are clear and the same across all ten languages, so we can be confident in the main findings.

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Appendix A. Code

The study was done using two Python scripts. **analyze.py** reads the human sentences from the UD database, works out the gap scores, and uses Stanza to label the AI sentences. **regression.py** runs the further study and draws Figures 3 and 4.

A.1 analyze.py

```
import stanza, conllu, numpy as np, pandas as pd
from scipy import stats

def sentence_stats(sentence):
    arcs = [abs(t['id'] - t['head']) for t in sentence
             if isinstance(t['id'], int) and t['head'] != 0]
    L = len([t for t in sentence if isinstance(t['id'], int)])
    adl = np.mean(arcs) if arcs else 0.0
```

```

ttr = len(set(t['form'].lower() for t in sentence
if isinstance(t['id'], int))) / max(L, 1)
return {'adl': adl, 'nadl': adl / np.sqrt(L), 'L': L, 'ttr': ttr}

# Load UD treebank (gold trees)
with open('en_ewt-ud-train.conllu') as f:
human_sents = conllu.parse(f.read())
human_df = pd.DataFrame([sentence_stats(s) for s in human_sents])
human_df['corpus'] = 'human'

# Parse LLM sentences with Stanza
nlp = stanza.Pipeline('en', processors='tokenize,pos,depparse')
records = []
for text in llm_sentences:
doc = nlp(text)
for sent in doc.sentences:
arcs = [abs(w.head - w.id) for w in sent.words if w.head > 0]
L = len(sent.words)
if arcs:
adl = np.mean(arcs)
ttr = len(set(w.text.lower() for w in sent.words)) / L
records.append({'adl': adl, 'nadl': adl/np.sqrt(L),
'L': L, 'ttr': ttr, 'corpus': 'llm'})
llm_df = pd.DataFrame(records)

# KS test on normalised ADL
ks, p = stats.ks_2samp(human_df.nadl, llm_df.nadl)
print(f'nADL KS={ks:.4f}, p={p:.4e}')

```

A.2 regression.py

```

import pandas as pd, statsmodels.formula.api as smf

combined = pd.concat([human_df, llm_df])

# OLS interaction model: ADL ~ sentence_length * corpus + TTR * corpus
model = smf.mixedlm(
'adl ~ L * corpus + ttr * corpus',
data=combined,
groups=combined['language'] # language as random intercept
).fit()
print(model.summary())

# Extract intercepts per corpus
b_human = model.params['Intercept']
b_llm = model.params['Intercept'] + model.params['corpus[T.llm]']
print(f'Human intercept: {b_human:.3f}')
print(f'LLM intercept: {b_llm:.3f}')

```